VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING



Machine Learning

Individual Assignment – Part I

Human Activity Recognition
Part I Portfolio Summary

Advisor(s): NGUYEN AN KHUONG
Student(s): PHAM MINH TRI 2353235

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1 Problem and Experimental Protocol

This report summarizes the pre-midterm portfolio for CO3117 Machine Learning[1]. The fixed use case is human activity recognition from smartphone inertial measurements. Across model families, the dataset, target and official held-out population remain unchanged. The purpose is to connect mathematical assumptions, implementation choices and observed model behavior under a common evaluation protocol, rather than optimize a leaderboard score.

1.1 Dataset and target

UCI Human Activity Recognition Using Smartphones (UCI HAR, version 1.0) contains 561 engineered features from accelerometer and gyroscope measurements[2]. The target has six classes: WALKING, WALKING UPSTAIRS, WALKING DOWNSTAIRS, SITTING, STANDING and LAYING. The official split provides 7,352 training windows from 21 subjects and 2,947 test windows from nine other subjects. Subject separation is important because nearby overlapping windows and person-specific movement patterns can make a random window split optimistic.

Table 1.1: Fixed protocol and internal validation population.

Item	**Configuration**
Dataset	UCI HAR v1.0; six-class activity classification
Official split	21 training subjects; nine test subjects
Internal fit	15 original training subjects; 5,270 windows
Internal validation	Six original training subjects; 2,082 windows
Validation subjects	1, 3, 8, 15, 25 and 27
Primary metric	Macro-F1 over the six actual activity labels
Secondary evidence	Accuracy, confusion matrices and runtime
Randomness	Split seed 42; paired MLP/GA seeds 42, 0 and 1

The internal split uses `GroupShuffleSplit` with a subject holdout fraction of 0.25 and seed 42. All comparisons in the main results table use this same validation population. MLP and Perceptron scaling parameters are estimated from the internal fit rows only. GA feature ranking and fitness use a further inner split of the fit subjects. No final-test data is used by the new Tree, MLP, GA or BN/TAN practice experiments.

1.2 Metric definition and evaluation correction

For each class k, let P_k and R_k denote precision and recall:

$$F_{1,k} = \frac{2P_kR_k}{P_k + R_k}, \qquad \text{Macro-F1} = \frac{1}{6}\sum_{k=1}^{6} F_{1,k}.$$

Undefined per-class ratios are assigned zero. Macro-F1 gives equal weight to each activity, while accuracy counts all correctly classified windows together. The earlier implementation averaged seven array entries, including an unused label 0. The 14 October correction explicitly uses HAR labels 1–6. The original results and the corrected reruns are documented separately in `results/PART1_METRIC_AUDIT.md`.

Test-set limitation. The original Perceptron learning-rate script repeatedly evaluated the official test set. This exposure cannot be reversed. Its corrected historical test scores are therefore reported separately as descriptive reruns, not as a fresh unbiased assessment. The internal validation comparison should also not be interpreted as a final test: it is reused for the prespecified model comparisons.

2 Model Families and Implementation Evidence

Table 2.1: Part I models and their implementation scope.

Family	**Scope**	**Main evidence**
Baselines	Baseline	Majority class and stratified random predictions
Decision Tree	Depth B	Attributed continuous-tree adaptation; stopping and pruning comparisons
Perceptron	Depth A core	Binary update and six-class one-vs-rest wrapper
MLP	Depth A core	NumPy forward/backward, gradient checks and capacity experiment
GaussianNB	Depth A core	Class statistics, log-posteriors and sklearn comparison
Genetic Algorithm	Depth B	Binary feature-mask search and elitism ablation
BN/TAN	Structural study	Factorization, d-separation and HAR suitability evidence

2.1 Decision Tree and Perceptron

The tree greedily maximizes information gain:

$$IG = H(S) - \frac{|S_L|}{|S|}H(S_L) - \frac{|S_R|}{|S|}H(S_R).$$

The adaptation retains the recursive splitting and majority-leaf structure of ML-From-Scratch[3]. It adds midpoint thresholds, vectorized cumulative class counts, a strict depth limit and a minimum child-size rule. Fit and prediction use the same $x_j \leq t$ routing convention. The existing impurity/split core is kept in `src/from_scratch/tree_split.py`.

For binary labels $y \in \{-1, +1\}$, the Perceptron update is

$$\text{if } y(w^\top x + b) \leq 0: \quad w \leftarrow w + \eta y x, \qquad b \leftarrow b + \eta y.$$

Six one-vs-rest classifiers extend this rule to HAR; prediction takes the class with the largest decision score. Separability gives a finite-convergence guarantee, but overlapping data need not converge within the epoch budget.

2.2 MLP and Gaussian Naive Bayes

The one-hidden-layer MLP computes

$$H = \tanh(XW_1 + b_1), \qquad P = \text{softmax}(HW_2 + b_2), \qquad L = -\frac{1}{n}\sum_{i,k} Y_{ik} \log P_{ik}.$$

Backpropagation uses $\Delta_2 = (P - Y)/n$ and $\Delta_1 = (\Delta_2 W_2^\top) \odot (1 - H^2)$, giving $\nabla W_2 = H^\top \Delta_2$ and $\nabla W_1 = X^\top \Delta_1$. The existing core implements these operations explicitly in NumPy. Provided contract tests compare its gradients with numerical differences. The added training wrapper uses mini-batch SGD rather than a high-level MLP estimator.

GaussianNB predicts the class maximizing

$$\log P(c) + \sum_j \log \mathcal{N}(x_j; \mu_{cj}, \sigma_{cj}^2 + \epsilon), \qquad \epsilon = 10^{-9} \max_j \text{Var}(X_{\cdot j}).$$

Means and population variances are estimated per class, while the smoothing scale uses all fit rows, including between-class variation[4]. Log-sum-exp normalizes the log-posteriors without multiplying many small likelihoods. Normalization does not, by itself, ensure probability calibration.

Attribution and verification. Exact upstream snapshots, their MIT license, content hashes and equation-to-code locators are saved in `docs/reference/`. Codex assisted with the new adapters, training wrapper, experiments, corrections and report drafting; the corresponding episodes and verification are recorded in `AI_USE.md`. The current automated suite reports 36 passed tests and 15 optional skips at W07. The optional components include models scheduled for later course weeks.

3 Comparable Validation Results

Table 3.1 summarizes the common subject-held-out validation results. The MLP rows average three paired training seeds; their standard deviations describe initialization/training variation on one fixed split, not population uncertainty. The GA row is the prespecified seed-42 configuration, not the configuration with the highest outer-validation score.

Table 3.1: Comparison on the same 2,082 validation windows.

Model/configuration	**Macro-F1**	**Accuracy**
Majority baseline	0.0520	0.1849
Stratified random, seed 42	0.1645	0.1676
Adapted Tree, unlimited	0.8624	0.8713
Adapted Tree, depth 6	0.8719	0.8833
sklearn Tree, unpruned	0.8815	0.8881
OvR Perceptron	0.9496	0.9544
MLP, width 32, three-seed mean	0.9495	0.9539
MLP, width 64, three-seed mean	0.9504	0.9549
GaussianNB, all 561 features	0.7224	0.7363
GaussianNB, 12 GA candidates	0.7407	0.7594
GA + GaussianNB, seven features, seed 42	0.7310	0.7522

The majority predictor always chooses LAYING, the most frequent fit class. Its low Macro-F1 exposes failure on the other five activities. The stratified random predictor is a stronger chance baseline but still ignores feature information. Every trained family substantially exceeds these baselines.



Perceptron and both MLP widths achieve similar validation performance on the engineered representation. The small mean MLP advantage is insufficient to establish a reliable ranking on this holdout. Trees perform less well but provide explicit split rules and a direct complexity analysis. GaussianNB is fast to fit, yet its stronger distributional and independence assumptions are associated with systematic errors and overconfident probabilities.

3.1 Historical test reruns: separate evidence

Table 3.2: 14 October reruns on the previously exposed official test set.

Fixed historical configuration	**Macro-F1**	**Accuracy**
Majority baseline	0.0514	0.1822
Stratified random, seed 42	0.1618	0.1639
OvR Perceptron, $\eta = 0.01$, 100 epochs	0.9545	0.9539

These rows must not be mixed with the validation table to claim a model ranking. The previously recorded Perceptron values were F1 0.8143 and accuracy 0.9494. The new F1 is not merely the old value multiplied by 7/6: accuracy also differs, and the original prediction vectors were not retained to explain the discrepancy. The current source was rerun with the fixed historical configuration, with the result explicitly dated and labeled as a rerun.

3.2 Perceptron sensitivity

Learning rates 0.001, 0.005, 0.01, 0.05 and 0.1 give identical validation scores in the new sweep. With zero initialization and a fixed positive step size, the update rule scales weights and biases together in exact arithmetic; changing that scale need not change prediction signs or one-vs-rest ranking. No learning-rate advantage is demonstrated. The first tied candidate is saved deterministically for validation; the historical test configuration remains unchanged.

4 Controlled Tree and MLP Experiments

Settings and expected effects were recorded before running in `docs/experiments/PART1_PLAN.md`. The comparisons hold the data population, metric and other settings fixed

within each experiment.

4.1 Tree complexity and pruning

Table 4.1: Stopping and pruning: generalization versus tree complexity.

Configuration	**Train F1**	**Val. F1**	**Depth**	**Leaves**
Adapted, unlimited	1.0000	0.8624	14	100
Adapted, depth 6	0.9529	0.8719	6	38
Adapted, minimum leaf size 20	0.9569	0.8546	11	54
sklearn, $\alpha = 0$	1.0000	0.8815	14	100
sklearn, $\alpha = 0.002$	0.9898	0.8706	12	65

The unlimited tree perfectly fits training labels but generalizes less well. Restricting the adapted tree to depth six removes 62 leaves and improves validation F1 by about 0.0095. This supports the expected bias–variance trade-off for that comparison. However, a minimum leaf size of 20 reduces validation performance, and sklearn post-pruning also lowers its score. Regularization must therefore be evaluated empirically rather than assumed to help.

The adapted tree's leading split-importance features are `tBodyAccJerk-std()-X`, `tGravityAcc-min()-X` and `fBodyAccMag-std()`. Upstream and adapted implementations may resolve equal-gain candidates or floating-point thresholds differently; exact HAR prediction agreement is not claimed. Known-split tests independently check midpoint routing, majority leaves and stopping behavior.

4.2 MLP capacity

Both MLPs use one tanh hidden layer, mini-batch SGD with learning rate 0.05, batch size 128 and 60 epochs. No early stopping or validation-selected epoch is used. Within a seed pair, sample shuffling is identical; hidden width is the only changed capacity setting.

Table 4.2: Validation F1 for paired hidden-width comparisons.

Training seed / summary	**32 units**	**64 units**
42	0.9522	0.9514
0	0.9563	0.9539
1	0.9400	0.9460
Mean	0.9495	0.9504
Sample standard deviation	0.0085	0.0040
Parameters	18,182	36,358
Mean fit time (seconds)	1.24	2.10

Increasing width improves training F1 in all three pairs but improves validation F1 only for seed 1. Its mean benefit is approximately 0.001, smaller than the observed seed variation. The larger network roughly doubles parameter count and increases runtime without a demonstrated robust generalization benefit. Loss histories also retain the training/validation gap rather than judging learning from the final training score alone.

The new paired runs use one BLAS thread. These machine-dependent SGD runtimes are not directly comparable to the separate historical HW3 sklearn/Adam runs. The new scripts provide reproducible evidence without replacing the handwritten historical submissions.

5 Genetic Search and Bayesian-Network Study

5.1 Bounded GA feature selection

The reference GA originally searches character strings [3]. The adaptation uses a 12-bit feature mask, tournament selection, single-point crossover and independent bit mutation. Empty masks are repaired; repeated masks reuse cached fitness. A second variant retains one unchanged elite per generation. Both variants return the best candidate seen over the whole run.

ANOVA ranking uses 3,911 windows from 11 inner-fit subjects. Fitness uses 1,359 windows from four separate inner-validation subjects, all within the outer fit population:

$$F(z) = \text{Macro-F1}_{\text{inner}}(z) - 0.02\frac{\|z\|_0}{12}.$$

The chosen mask is refit with GaussianNB on all 15 outer-fit subjects before evaluation on the six outer-validation subjects. Thus neither ranking nor search fitness uses outer-validation labels.

Population size is 16; the budget is 12 evaluated generations; mutation probability is 1/12 per bit. Seeds 42, 0 and 1 are paired across elite counts, with identical initial populations. The focused variable is elite retention.

Table 5.1: GA outcomes: inner fitness and outer validation are distinct.

Seed	**Elites**	**Inner fitness**	**Features**	**Outer F1**
42	0 / 1	0.674613	7	0.731027
0	0 / 1	0.674613	7	0.731027
1	0	0.674613	7	0.731027
1	1	0.672145	8	0.740619
All 12 candidates	–	–	12	0.740683

Elitism prevents the current-generation best fitness from falling, but does not guarantee finding a better optimum within the budget. Seed 1 with one elite has lower inner fitness yet better outer F1 than the seven-feature mask. This illustrates imperfect transfer of fitness ranking across subject groups. The main benefit here is feature compression, not an accuracy improvement over the all-12 baseline. Three seeds and a restricted candidate pool do not prove global optimality. Fitness and normalized pairwise Hamming diversity are retained for every evaluated generation.

5.2 BN/TAN factorization and suitability

For the illustrative graph $A \to C \leftarrow B$, $C \to D$, $B \to E$:

$$P(A, B, C, D, E) = P(A)P(B)P(C \mid A, B)P(D \mid C)P(E \mid B).$$

The binary graph needs $1+1+4+2+2 = 10$ independent parameters. Enumeration over 32 states verifies normalization and the four worked d-separation cases. The unobserved collider blocks the A–B path; observing C or its descendant D activates it. Conditioning on C blocks the chain from A to D. This is a probability illustration, not another predictive benchmark dataset.

Naive Bayes factors $P(Y, \mathbf{X}) = P(Y) \prod_j P(X_j \mid Y)$. A three-feature TAN can instead

use

$$P(Y)P(X_1 \mid Y)P(X_2 \mid Y, X_1)P(X_3 \mid Y, X_2).$$

On HAR fit rows, magnitude mean and energy have within-class Pearson correlations between 0.868 and 0.988, challenging strict conditional independence for that pair. TAN could represent some such dependence, but continuous conditional-density estimation or fit-only discretization would be required. This portfolio includes a structural suitability study, not a fitted TAN classifier or a claimed TAN performance gain.

6 Error Analysis, Synthesis and Reproducibility

6.1 Systematic error patterns

Full six-class confusion matrices are saved for each run. GaussianNB's largest directed validation error is STANDING predicted as SITTING: 264 of 375 standing windows. Its mean maximum posterior is 0.9971, despite accuracy 0.7363; clipped multiclass log loss is 7.1509. This indicates overconfidence. Correlated features are a plausible contributor, although this experiment does not establish a causal explanation.

For the adapted depth-six tree, 63 WALKING DOWNSTAIRS windows are predicted as WALKING UPSTAIRS. The same confusion is the largest directed error in every MLP run, with 33–50 cases across configurations. The seed-42 seven-feature GA classifier instead confuses 189 SITTING windows with STANDING. Feature compression therefore changes error structure as well as model size.

6.2 Cross-model interpretation and limits

The Perceptron supplies a strong linear discriminative baseline; MLP adds nonlinear capacity, while GaussianNB models class-conditional densities. Trees construct axis-aligned partitions, and GA searches a representation rather than serving as an activity classifier by itself. On this engineered feature set, extra complexity does not consistently improve validation. Model assumptions, fit behavior, class-specific errors and computational cost are more informative than one headline score.

The study is limited by one outer subject split, three training seeds for MLP/GA, bounded GA search and prior exposure of the official test set. Seed variability should not be presented as a confidence interval across unseen people. New experiments and corrections are dated 14 October; historical weekly checkpoints and scans remain separate

evidence. The current handwritten A4 notes are an interim revision, with their preparation status recorded in `SUBMISSION_PART1.md`.

6.3 Reproduction

The tested environment is Python 3.13.13, NumPy 2.5.1 and scikit-learn 1.9.1. Exact package versions are recorded in `requirements-part1-lock.txt`. Run from the repository root with the frozen HAR data already obtained:

```
python experiments/part1_pre_midterm/w05_baseline.py
python experiments/part1_pre_midterm/w05_perceptron.py
python experiments/part1_pre_midterm/w06_naive_bayes.py
python experiments/part1_pre_midterm/w05_tree.py
python experiments/part1_pre_midterm/w05_mlp.py
python experiments/part1_pre_midterm/w07_ga.py
python experiments/part1_pre_midterm/w07_bn.py
```

These default commands evaluate internal validation only. The two W05 `--evaluate-test` options are reserved for explicitly labeled historical test reruns. `results/validation_metrics.csv`, audit JSON files and `docs/reference/DISSECTION.md` provide scores, subject IDs, settings, diagnostics, exact source hashes and equation-to-function mapping. The new experiment scripts, checks and this report were prepared with Codex assistance, as recorded in `AI_USE.md`.

References

[1] CO3117 Machine Learning, HK261. *Individual Longitudinal Assignment: One Dataset, One Use Case, Many Models*. Course specification, Sections 4, 7 and 9–11.

[2] D. Anguita, A. Ghio, L. Oneto, X. Parra and J. L. Reyes-Ortiz. *A Public Domain Dataset for Human Activity Recognition Using Smartphones*. ESANN, 2013. Dataset: UCI HAR, version 1.0, UCI dataset identifier 240.

[3] E. Linder-Norén. *ML-From-Scratch*, classification-tree and genetic-algorithm implementations. MIT-licensed snapshots retrieved on 14 October 2026; paths and SHA-256 hashes in `docs/reference/sources.json`.

[4] *scikit-learn*, installed version 1.9.1. GaussianNB, DecisionTreeClassifier, GroupShuffleSplit and metric implementations. Relevant functions and comparison procedures are documented in `REFERENCES.md`.